NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

Mathematics II

T210

Tuesday, 30/07/2024

08:30 – 11:30 AM

Names:.....

Index number :.....

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2023-2024

QUESTIONS and ANSWERS BOOKLET

OPTIONS/ TRADES: Accounting (ACC), Business Services (BUS), Graphic Arts (ART), Sculpture and Ceramics (SCE), Tailoring (TAL), Leather Works (LTW), Music (MUS), Culinary Arts (CUA), Food and Beverage Services (FBS), Front Office Operations (FOO), Tourism (TRS), and Office management (OFM).

SUBJECT/EXAM: MATHEMATICS II

DURATION: 3 Hours

Read carefully the instructions on page 1 & 2.

TOTAL MARKS...../100

FOR EXAMINER'S USE ONLY

QUESTIONS	1	2	3	4	5	6	7	8	9	10	Total
Marks											

QUESTIONS	11	12	13	14	15	16	17	18	19	20	Total
Marks											

QUESTIONS	21	22	23	24	25	26	27	28	29	30	Total
Marks											

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2023-2024

INSTRUCTIONS TO CANDIDATES (ANSWER BOOKLET)

1. A candidate should fill in the actual names and the Index number on the cover of this questions and answer booklet on the provided place.

2. It is illegal for a candidate to write any of names, Index number or school name inside the answer booklet.

3. No candidate should remove or tear any pages or part of it in the answer booklet.

4. A candidate should answer in the language in which the examination is set.

5. A candidate should sign on the sitting plan when submitting the answer booklet. He/she has also to check if the answer booklet is well sealed.

6. No extra paper is allowed in the examinations room. If a candidate is caught with it his/her results will be nullified.

7. No candidate is allowed to write answers not related to the subject being sat for, otherwise it will be considered as a cheating case.

8. Write your answers on the 16 lined pages (From page 7 to page 22).

9. Use the last non-lined pages as draft.

10. Results for any candidate who is caught in examination malpractices are nullified. The cheating can be recognized during examinations administration, marking exercise or even thereafter.

N.B: 1) After results publication, there is no remarking and no candidate is given his/her answer booklet for review. This answer booklet is a property of NESA.

2) Claims are only received online within 30 days after results publication. A link will be provided after results publication.

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2023-2024

OPTIONS/TRADES: Accounting (ACC), Business Services (BUS), Graphic Arts (ART), Sculpture and Ceramics (SCE), Tailoring (TAL), Leather Works (LTW), Music (MUS), Culinary Arts (CUA), Food and Beverage Services (FBS), Front Office Operations (FOO), Tourism (TRS), and Office management (OFM).

SUBJECT/EXAM: MATHEMATICS II

DURATION: 3 HOURS

INSTRUCTIONS TO CANDIDATES (QUESTION PAPER)

This Exam paper is composed of Two Sections (A and B). Follow the instructions given below, and answer the indicated questions for a total of 100 marks

Section **A**: Fifteen (**15**) questions, all **Compulsory** **55marks**

Section **B**: Among the six (**6**) questions, attempt any three (3) **45 marks**

Allowed materials:

- Geometrical Instruments
- Silent non-programmable calculators
- Blue and black pen only.

Note:

Every candidate is required to carefully comply with the provided assessment instructions.

T 210_ Mathematics II

SECTION A: Attempt all questions **(55 marks)**

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01. Binary operator $T$is defined in the set of all natural numbers by **(2 marks)**

$$xTy = x^y + y^x$$

a) Calculate $2T3$

b) Calculate 3T2

02. Solve the following system by substitution **(4 marks)**

$$\begin{cases} 3x - 5y = 23 \\ x - 4y = 3 \end{cases}$$

03. Given the complex number $Z = i$, calculate the values of: **(3 marks)**

a) Z^2

b) Z^5

04. Find the median, mode and range of: **(3 marks)**

6, 2, 8, 12, 3, 5 ,20, 15, 3

05. Using factorization method, evaluate the limit: $\lim_{x \to 2} \frac{x^4 - 16}{x^2 - 4}$ **(4 marks)**

06. Find the relative asymptotes to the curve representing **(5 marks)**

$$f(x) = \frac{x^2 + 3x - 1}{3x^2 - 27}$$

07. Given the function $f(x) = x^{13} + 2x$,find: **(3 marks)**

a) $f'(x)$

b) $f'(1)$

08. Solve the equation: $2\log_2 x + \log_x 2 = 3$ **(5 marks)**

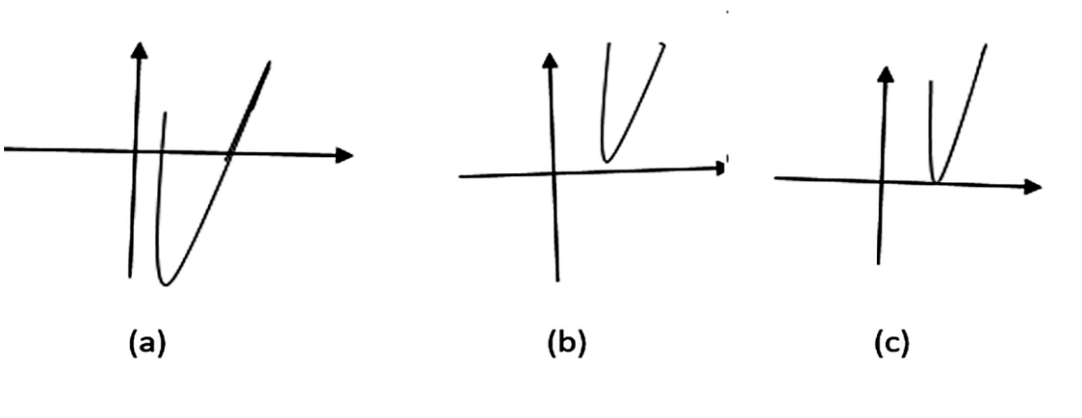
09. Solve in $\mathbb{R}$: $4^{3x+1} = 64$ **(3 marks)**

10. Given that $\log 2 = 0.30$, $\log 3 = 0.48$ and $\log 5 = 0.69$ without using calculator, calculate: a) $\log 0.9$ **(5 marks)**

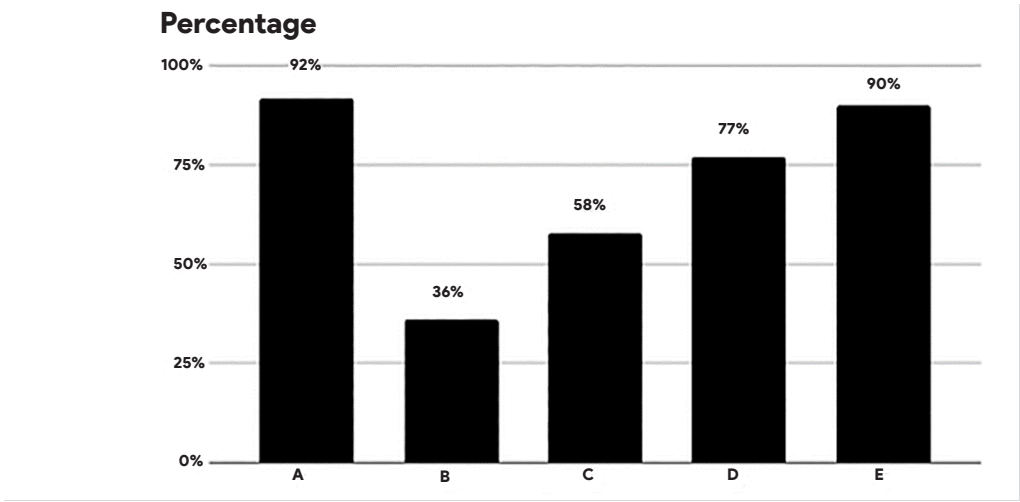
b) $\log 150$

11. In how many ways can three cars be parked in a parking with four places? **(2 marks)**

12. Among the following graph, specify the graph which has: two equal roots, two distinct roots, no real roots. **(3 marks)**



13. a) The graph below is the simple vertical bar chart of marks in percentages of five students: **A, B, C, D** and **E** from a certain class. **(5 marks)**



i. What is the name of student who has got maximum marks in this class?

ii. What is the name of student who has got 58% marks in this class?

iii. What is the value of minimum marks in this class?

b) i. Write 22% as a fraction

ii. Write 34% as decimal.

14. Determine whether, the identity is true or false, if false find appropriate equivalent expression. **(4 marks)**

$$sec\,\theta\sqrt{1-sin^2\,\theta}=1$$

15. Find the real part, argument and the module of the following complex number **(4 marks)**

$$Z=\frac{1}{2}+\frac{\sqrt{3}}{2}i$$

Section B: Attempt any three (3) questions **(45 marks)**

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16. Solve using the Cramer's rule **(15 marks)**

$$\begin{cases}3x+2y-5z=2\\ x+2y=3\\ 2x-y+z=-3\end{cases}$$

17. If A is an angle in Quadrant 1 and B is an angle in Quadrant 2 such that $\cos A=\frac{3}{5}$ and $\sin B=\frac{3}{5}$, evaluate each of the following without using a calculator: **(15 marks)**

a) a. $\cos(A+B)$

b) $\sin(A-B)$

c) $\cos(A-B)$

18. Given the function $f(x)=\frac{x+1}{(x-3)^2}$. determine: **(15 marks)**

a) The domain of $f(x)$.

b) The x-intercept(s) and y-intercept(s).

c) All asymptotes to the curve of the function $f(x)$.

d) The first derivative of $f(x)$.

e) The Interval on which $f(x)$ is increasing or decreasing.

f) Sketch the graph of $f(x)$ in a Cartesian plane

19. Given the data 2, 4, 5, 8, 11. Determine the**:** **(15 marks)**

a) Mean.

b) Median.

c) Mode.

d) Range.

e) Variance.

f) Standard deviation

20. Given two matrices as $A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \\ 0 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 3 \\ -1 & 0 & 1 \end{bmatrix}$. **(15 marks)**

Compute:

a) $A - B$

b) $2A + B$

c) $A^T - 4B$, where A^T stands for the transpose of the matrix A

21. **a)** Evaluate: $\frac{5}{6} + \frac{1}{2} \times \frac{3}{7} \div \frac{1}{3}$. **(15 marks)**

b) Express 0.35 as a percentage.

c) Convert $\frac{3}{4}$ into a percentage.

END OF ASSESSMENT

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